Log Analysis

Service Logs and Event Logs & Some Packet

What Happened?

P1KA-S4UR

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Created by	Jerome Gutierrez
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Overview

1. What is Log Analysis
2. Log Analysis of Different Services

Log Analysis

Definition and Challenges

What is Log Analysis?

- Analyzing the various *log files* in order to **identify anomalies**, **provide statistics**, and **to aid baselining**

- It is also about identifying the types of data you can get from log files and how you can correlate them to other data sources in order to paint a clear picture of what transpired

Challenges of Log Analysis

- Has the following challenges:
  - **Volume**
    - A lot of logs are being generated every second and scales with organization size
  - **Different file formats** for **different applications** on essentially the same data sources
    - Ex. Multiple syslog formats
  - **Mostly comprised of noise**
    - Most of the logging that are enabled in the data sources are turned on for compliance purposes and not inherently implemented with security in mind

Steps of Log Analysis

Collection > Enrichment > Analysis

Log Analysis Steps

- **Data Collection**
  - Involves the collection and parsing of data from various log sources
- **Data Enrichment**
  - Involves improving the data that was collected
  - Example
    - WhoIS lookup for IP addresses
    - Reputation lookup for Domain names and IPs
- **Data Analysis**
  - Involves making inferences/correlations based on the data that was processed and indexed in the log collection tool / SIEM (System/Security Information and Event Management)

Services

Logs of Services

Bringing Things Into Focus: Services

- SMTP
- DNS
- HTTP
- HTTPS

Log Analysis: SMTP

- **SMTP**
  - One of the most common entry points for attacks
  - Attackers often use this service to conduct
    - Phishing attacks
    - Click-through malware

Log Analysis: SMTP

- **SMTP**
  - Common SMTP Fields
    - Source IP
    - Destination IP
    - File Attachment Name
    - File Attachment Size

Log Analysis: SMTP

- **SMTP**
  - Common SMTP Fields
    - HELO
    - From
    - To
    - Subject
    - Is Webmail
    - First Received
    - Reply Codes
    - Mail User Agent (MUA)

Log Analysis: SMTP

- **SMTP**
  - Valuable (Value Add) SMTP Fields
    - Display Names (from users)
    - Geological IP
    - From and To Domains

Log Analysis: SMTP Tips

- **SMTP**
  - Email Monitoring
    - Inbound Monitoring
      - Fuzzy Searching
      - Bursts from outside
      - Key Employee Names
    - Outbound Monitoring
      - Monitoring of authorized sources
      - Monitoring of SMTP MUA (Mail User Agent)

Log Analysis: SMTP Tips

- **SMTP**
  - Email Monitoring
    - Inbound Monitoring
      - Fuzzy Searching
      - Take legitimate domains and look for variants
        - Original
          - securebank.com.ph
        - Derivatives/Variants
          - securbank.com.ph
          - securbaank.com.ph
          - secure.bank.ph

Log Analysis: SMTP Tips

- **SMTP**
  - Email Monitoring
    - Inbound Monitoring
      - Bursts from outside
      - Phishing can involve e-mailing large amounts of internal employees
      - Note:
        - Requires filtering out noise(e.g., Marketing)
        - Requires threshold

Log Analysis: SMTP Tips

- **SMTP**
  - Email Monitoring
    - Inbound Monitoring
      - Key Employee Names
        - Use a list of names to look for phishing attempts
          - Look for display names of Key Staff

Log Analysis: SMTP Tips

- **SMTP**
  - Email Monitoring
    - Outbound Monitoring
      - Monitoring of authorized sources
      - Normally, email should come from authorized servers or subnets
        - Exceptions may be scripts that send e-mails from hosts
      - Filter out
        - Email addresses used by scripts
        - Authorized Sources (w/ Business Justification)
      - Investigate outbound emails from unauthorized IPs

Log Analysis: SMTP Tips

- **SMTP**
  - Email Monitoring
    - Outbound Monitoring
      - Monitoring of SMTP MUA (Mail User Agent)
      - Logs contain an SMUA field
        - Specifies what client application sends an email
      - Can help filter out authorized outbound emails
        - Great in combination with IP filters
        - Can also help find source of infections

Log Analysis: SMTP Tips

- **SMTP**
  - Email Monitoring
    - Outbound Monitoring (Additional)
      - Outbound SMTP Clipping Level
        - What is the max emails sent per system, per hour?
        - What if that was to increase exponentially?
        - Can be legitimate increase
        - Could be operational issue
        - Could be SMTP command and control
        - Could be phishing attacks using authorized system
      - Often, malware attacks companies through another victim company

Log Analysis: DNS

- **DNS**
  - Common DNS Fields
  - Answer
  - Request
  - Query Class
  - Query Type
  - Response Codes
  - TTLs

Log Analysis: DNS

- **DNS**
  - Valuable (Value Add) DNS Fields
    - Frequency Score (detect DGAs)
    - Parent Domain (google.com)
    - Top Level Domain (com)
    - Subdomain (www)
    - Domain Lengths
    - Domain Age
  - Optional Additions
    - Geo-information
    - Tagging

Log Analysis: DNS Tips

- **DNS**
  - Some useful tools
    - DNS Sinkhole
      - Jason Fossen developed a PowerShell sinkhole script
        - Takes known bad domains and uses DNS to prevent access
        - Download it at: https://blueteampowershell.com/

Log Analysis: DNS Tips

- **DNS**
  - Some useful tools
    - Freq_server.py and freq.py
      - Randomness calculation against a string to determine if it is “random” i.e., generated by an algorithm rather than a human
        - API-based
        - You may opt to use logstash REST plugin to enrich your logs

Log Analysis: DNS Tips

- **DNS**
  - Some useful tools
    - domain_stats.py
      - Provides creation_date, Alexa top 1m and Cisco top 1m lookups
        - API-based
        - Enables you to tag a domain depending on the various attributes that it has